

BIOT 6900 · Module 2: Finding Drug Targets for Alzheimer's Disease

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Key terms

- **RNA:** how much a gene is being “read” in a tissue.
- **Protein:** how much of the gene's final product is present.
- **GWAS:** studies that find DNA variants linked to a disease.
- **Fold change (lfc):** how much a gene changes in AD vs. healthy brains. Above 0 = up, below 0 = down.
- **−log10 p:** how strong a genetic link is. Bigger = stronger.
- **Concordant:** RNA and protein move in the same direction.

1. Data

I found and downloaded all the data myself with one script (`scripts/get_data.py`). All sources are free, with no login or application needed.

Layer	Source	Access	Genes
RNA	Agora (agora.adknowledgeportal.org)	Open	20,539
Protein	Agora (agora.adknowledgeportal.org)	Open	8,254
Genetics	GWAS Catalog (ebi.ac.uk/gwas)	Open	3,558

- **RNA:** AD vs. healthy brains, from 3 donor groups (ROSMAP, Mayo, MSBB) and 9 brain regions.
- **Protein:** AD vs. healthy brains, from the prefrontal cortex.
- **Genetics:** every published DNA variant linked to Alzheimer's.

The data is unmatched. Each file gives one number per gene, not per person, so I joined the layers by gene name.

2. Steps

1. Load the three files.
2. For RNA, keep each gene's biggest change across brain regions (skipping the cerebellum, which Alzheimer's mostly spares).
3. Keep only genes found in all three layers. **1,115 genes** made it.
4. Mark a gene **concordant** if RNA and protein move the same way. **635 of 1,115 (57%)** do.
5. Rank each layer from 0 to 1 and add the three ranks with equal weight (1/3 each).
6. Sort by score. Save the top 15 to `data/targets_ad.csv` and all 1,115 to `data/targets_ad_full.csv`.

3. Results

Top 15 genes

Rank	Gene	RNA	Protein	Genetics	Concordant	Score
1	HLA-DRA	0.80	0.14	16.7	Yes	0.957
2	HLA-DRB1	0.66	0.10	22.5	Yes	0.955
3	APOC1	0.51	−0.23	672.7	No	0.950
4	PECAM1	0.86	0.09	16.5	Yes	0.939
5	MTSS2	0.64	−0.10	13.5	No	0.913
6	ITGAX	0.70	0.12	11.7	Yes	0.912

Rank	Gene	RNA	Protein	Genetics	Concordant	Score
7	GMPT	1.04	0.23	10.1	Yes	0.910
8	RPH3A	-1.16	-0.13	10.0	Yes	0.905
9	NRXN3	-0.64	0.06	22.0	No	0.904
10	APLN	1.28	0.10	10.5	Yes	0.902
11	APOE	0.38	0.11	496.4	Yes	0.902
12	SHANK2	-0.55	-0.06	30.0	Yes	0.899
13	APOB	0.40	0.13	19.0	Yes	0.897
14	MOG	0.47	-0.09	17.3	No	0.896
15	PLEC	0.60	0.26	10.5	Yes	0.895

RNA and Protein = fold change. Genetics = $-\log_{10} p$.

4. What the results mean

Why Alzheimer's and why this data

- Alzheimer's is the most common cause of dementia and has very few treatments.
- It follows on from Module 1, which used the APOE variant rs7412.
- Agora combines 3 donor groups and 9 brain regions, so it is bigger than a single GEO study.
- The GWAS Catalog collects every published Alzheimer's genetics result.

Do the top genes make sense?

Yes. Known Alzheimer's risk genes show up:

Gene	Rank	Note
APOE	11	Biggest known genetic risk for Alzheimer's. Up at RNA and protein.
HLA-DRB1	2	Known risk gene. Immune system.
CLU	31	Known risk gene. Fat transport.
BIN1	53	Known risk gene.
PICALM	83	Known risk gene.
SORL1	220	Known risk gene.

The top 15 genes fall into four groups that are all known to change in Alzheimer's:

1. **Brain immune cells:** HLA-DRA, HLA-DRB1, ITGAX (all up).
2. **Fat transport:** APOC1, APOE, APOB.
3. **Synapses (nerve connections):** RPH3A, NRXN3, SHANK2 (all down, which fits synapse loss).
4. **Blood vessels:** PECAM1, APLN.

Surprise hit: GMPT (rank 7) GMPT is not a known Alzheimer's gene, but it goes up at both RNA (+1.04) and protein (+0.23), and both results are strong. That makes it worth a closer look.

Missing genes TREM2, CD33 and MS4A6A are known Alzheimer's genes but are not in the list. The protein study did not measure them, so they were dropped in step 3.

A warning about APOC1 APOC1 (rank 3) sits right next to APOE on chromosome 19. Its huge genetic score (673) is probably mostly APOE's signal, not its own.

A gene where RNA and protein disagree: APOC1

	Change	Brain region
RNA	up (+0.51)	parahippocampal gyrus
Protein	down (−0.23)	prefrontal cortex

Both changes are statistically strong. Possible reasons:

1. RNA and protein were measured in **different brain regions**.
2. APOC1 is a protein that cells **send out**, so less stays in the tissue.
3. The protein may be **broken down faster** or **trapped in plaques**.
4. There may be **more immune cells** in the tissue, raising RNA without each cell making more.

Not every “No” is real. BIN1 is marked not concordant, but its protein change is tiny and not significant ($p = 0.43$), so that is probably just noise.

What we can say

- These genes have several separate kinds of evidence (RNA, protein and genetics).
- The three layers barely overlap, so a high score is not one signal counted three times.

What we cannot say

1. We can't say anything per patient. In Part 1 (breast cancer), RNA and protein came from the same tumors, so we could compare them person by person. Here the data is unmatched.
2. We can't say what causes what. A genetic link points to a region of DNA, not a proven cause.
3. We can't say the missing genes don't matter. TREM2 was dropped only because protein wasn't measured.
4. We can't give exact effect sizes. Taking the biggest RNA change per gene makes RNA effects look larger. Only 15% of genes have a significant protein change.

Weighting

I kept equal weights (1/3 each). Ranking each layer first stops APOE's huge genetic score from taking over. Protein could get less weight because its changes are small, but I kept equal weights to match the course default.

5. Conclusion

TODO (you): 2 to 3 sentences on what the list is useful for in Week 3, and what you would change.